Answer Key - Computer Science - Undergraduate Level Assessment 11 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) O(log e^N); B) O(log N); C) O(log log N); D) O(N)
   Note: Correct option: C - O(log log N)

2. **Answer:** C
   Options: A) It was first proven by Alan Turing.; B) It has not yet been proven, but finding a proof is a subject of active research.; C) It can never be proven.; D) It is now in doubt because of the advent of parallel computers.
   Note: Correct option: C - It can never be proven.

3. **Answer:** D
   Options: A) 53; B) 29; C) 50; D) 100
   Note: Correct option: D - 100

4. **Answer:** C
   Options: A) Error; B) abc; C) cba; D) c
   Note: Correct option: C - cba

5. **Answer:** A
   Options: A) a[i] == max; B) a[i] != max; C) a[i] < max || a[i] > max; D) FALSE
   Note: Correct option: A - a[i] == max

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'Chemistry'.

7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

8. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'Reference counting is well suited for reclaiming cyclic structures.'.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** False
    Options: A) True; B) False
    Note: LLM-generated false statement. Correct answer: 'Recursive procedures'.

Long Form Response

11. **Answer:** def increasing_trend(nums): | return True | return False
    Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** def check_none(test_tup): | return (res)
    Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key